

Machine learning analysis of electric arc furnace process for the evaluation of energy efficiency parameters

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HIGHLIGHTS

- The electric arc furnace process was analyzed using machine learning methods.
- The data-centric approach improved performance on test sets for four models.
- Quantitative and qualitative correlations between parameters and target are shown.
- An increase in natural gas decreases electricity consumption up to 40 kWh/t.
- Electricity consumption increases by about 1.5 kWh/t per operating minute.

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ABSTRACT

The electric arc furnace has been the subject of extensive research due to its complex and chaotic nature. Machine learning methods provide a powerful forensic examination of industrial processes as they exclude numerous assumptions and involve irregularities present in industrial conditions. In this study, different machine learning and data processing methods were used to evaluate the energy efficiency parameters of the electric arc furnace process. The dataset was collected over five years, in a steelmaking factory, with 42 features. This data was split into training and test sets, which were used for training and evaluation, respectively. With extensive data management, the data quality and machine learning model performance were improved. It was found that selected models display similar performance, yet the artificial neural network shows greater flexibility when changing targets. The results indicate that a data-centric rather than model-centric approach is better for improving model performance. Using the partial dependence plot and SHAP method, insight was gained into each parameter's correlation with the target. It was found that the amount of hot heel (melted steel left in the furnace, to be re-heated) was the main factor disturbing the data quality and model performance. It was also demonstrated that data for total oxygen consumption should be divided from the oxygen used into refining and natural gas. This highly improves model performance. Employing a data-centric machine learning model to control and optimize main process parameters (with a small capital investment) leads to lower energy consumption for industrial processes.

1. Introduction

In secondary steelmaking processes, a central place is reserved for an electric arc furnace (EAF). For many decades, the main objective of steel production in the EAF has been to improve energy consumption [1]. By reducing tap-to-tap time, electrode consumption, or exploiting chemical energy the electric arc furnace could reach its highest performance level. The optimization problem is nonlinear and complex, involving

numerous correlated parameters that impact the overall electrical consumption (Table 1). In most cases, an electric arc is automated. However, the control and optimization of other parameters rely on operator experience, which is valuable yet limited as it is impossible to comprehend multivariable interactions [2,3]. The problem of energy optimization often involves conflicting parameters, such as steel loss and CO₂ emissions. Thus, balancing these parameters requires making a lot of compromises and fine-tuning the process [4]. Making up approximately

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Table 1

Improvement of parameters that affect the energy consumption in the arc furnace.

Parameter	Improvement, optimization, or predicting approach	Reference
Scrap charging strategy	Dynamic control	[3,6]
	Linear quadratic regulation control design	[2]
Charging chemical composition	Exergy-economic approach	[7]
	Artificial neural network	[8]
Scrap type, shape, density	Statistical models, Random Forests	[9,10]
Maintenance and other technological delays	Linear regression model, genetic programming model	[6]
The profitability of the heat (batch)	Dynamic optimization, differential equations, algebraic model	[12]
Waste heat recovery, preheating	Advanced Process Simulation Software	[13]
	Physico-chemical modeling	[14]
Power quality optimization	Physico-chemical modeling	[15,16]
Electric arc length	Artificial Neural Networks	[17,18]
Water cooling system	Numerical simulation, experimental	[19]
Tap-to-tap time	Statistical models	[9,10]
Oxygen injection	Numerical methods, dynamic modeling and prediction, 3D simulation	[19–23]

20% of the annual industrial energy consumption, the iron and steel industry is responsible for 7–9% of the world's total CO₂ emissions, having a high for environmental impact [5]. New initiatives, such as the Industry 4.0 concept, aim to improve the sustainability of energy consumption in complex processes and to enforce greater control over hazardous pollutants. The most significant driving force of the Industry 4.0 concept is the machine learning approach applied to such large systems.

Although there are reasonably accurate physico-chemical models available, which simulate and optimize the EAF smelting process, they rely on simplifications and assumptions that do not take into account problems occurring in real working conditions [16,24,25]. In contrast, statistical models work with real-time data, enabling them to avoid these assumptions and simplifications related to the extraordinarily complex and dynamic behavior of scrap melting [9–11,26]. The main advantage of statistical modeling is its comprehensive analysis, which includes all parameters affecting the process. On the other hand, one disadvantage of statistical modeling is its data quality [11]. Over the past decades, machine learning methods have developed a great deal, their applications mainly focused on image, text, and sound processing. However, the research into machine learning applications for industrial processes still requires improvement, especially in the field of data processing and management.

This study aims to gain further insight into the real-time data of melting steel scrap in EAF, and to find the relation between the most influential parameters and electricity consumption, using data processing methods and machine learning analysis. The primary goal is to evaluate the energy efficiency parameters of the EAF process, using a data-centric approach and various machine learning techniques. To support this goal, the development of a data flow algorithm for data cleaning and feature selection was planned, increasing the machine learning model's performance. A regression analysis is carried out, and the outcomes of several machine learning models are compared: Support vector machine [27], Random Forest [28], XGBoost [29], and Artificial Neural Network (ANN) [30]. As opposed to prior studies, different targets are investigated: electricity consumption, operational time, and metallic steel yield. This will obtain a clear perspective on the impact of process parameters. Diverse views on different targets improve a model's transparency, along with the quantitative relationships of each

feature with the observed target. Data quality is a central problem of model performance, and this bad data arises from various issues under industrial conditions. Extensive data management is undertaken (removing outliers, combining domain knowledge, and statistical analysis) in order to improve data quality as well as machine learning model performance. All methods are fitted to the training data, and the model performance is checked via previously unseen test data. It is essential to divide the dataset into test sets, to gain experimental confirmation of model performances. The performance of each model's predictions is evaluated using the following metrics: coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE) metrics, maximum absolute error (MAX), and mean absolute percentage error (MAPE).

This paper aims to advance research in the field as follows:

- It shows that pointing to different targets, rather than simply energy consumption, has numerous benefits for analyzing complex processes, such as EAF. Targeting the yield of metallic steel and operating time reveals hidden interconnections of single parameters, e.g., oxygen consumption on energy consumption. Several different machine learning methods were used to compare which methods are most flexible when changing targets. Good forecasting performance was found for all methods, on a relatively small dataset, however ANN shows the most flexibility. Furthermore, research to date focuses only on parameters directly related to the process, while this study indicates that indirect measurements or parameters are important for improving models based on machine learning. For example, the impact of ladle furnace (LF) operating time is specified as being a relatively strong influence on all targets. Although LF is an entirely separate process from EAF, the effects of steel chemistry connects these two processes.
- A data-centric approach was introduced, along with an improved and effective method for extensive data management and cleaning. Although the proposed method is rigorous and removes a lot of data from the dataset, it has proven to bring high performance to prediction models. Other studies in the literature reviewed, which also use machine learning for process optimization of EAF, remove very little data, relying solely on expert analysis. Combining the visualization of parameter interdependence with domain knowledge leads to much better results.
- Unlike other research that deals with EAF using machine learning methods, this study examines the quantitative relationship between process parameters and different targets. The SHAP (SHapley Additive exPlanations) approach is proven to be powerful in relating the quantitative dependence of the target to one or two parameters (2D plot). This quantitative relationship has proven to be an advantage for optimizing a nonlinear process, such as EAF.

The rest of this paper is organized as follows: [Section 2](#) defines this study's experimental process in detail; [Section 2.1](#) describes the melting process in EAF; [Section 2.2](#) shows data management and cleaning methods, and [Section 2.3](#) displays the machine learning methods used. [Section 3](#) discusses the experimental results and analysis of the study's main contributions; [Section 3.1](#) explains the advantage of using different targets in the proposed methods and the evaluation of the model performance; [Section 3.2](#) demonstrates the quantitative relation between process parameters and selected targets, and [Section 3.3](#) shows the application options for the presented methods. Conclusions are drawn in [Section 4](#).

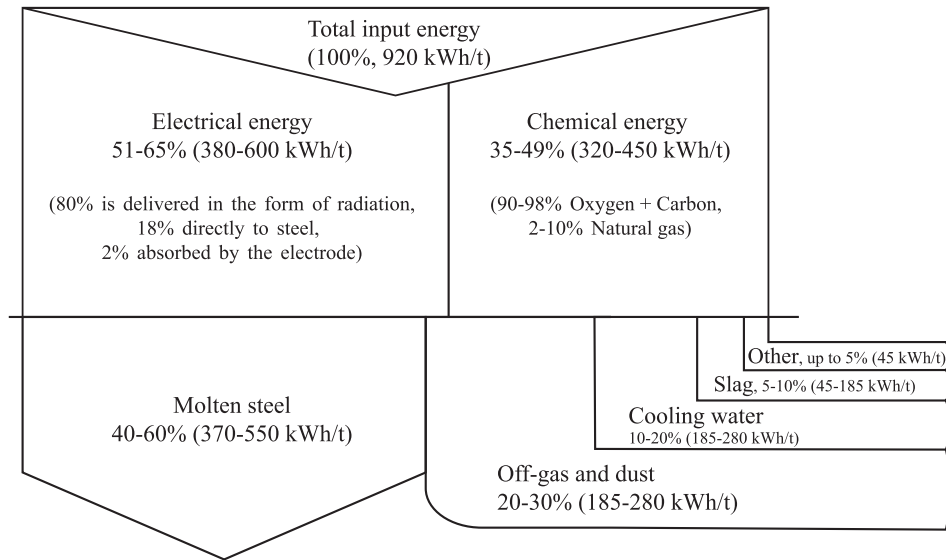


Fig. 1. The Sankey diagram for the electric arc furnace process describes the distribution and balance of energy consumption.

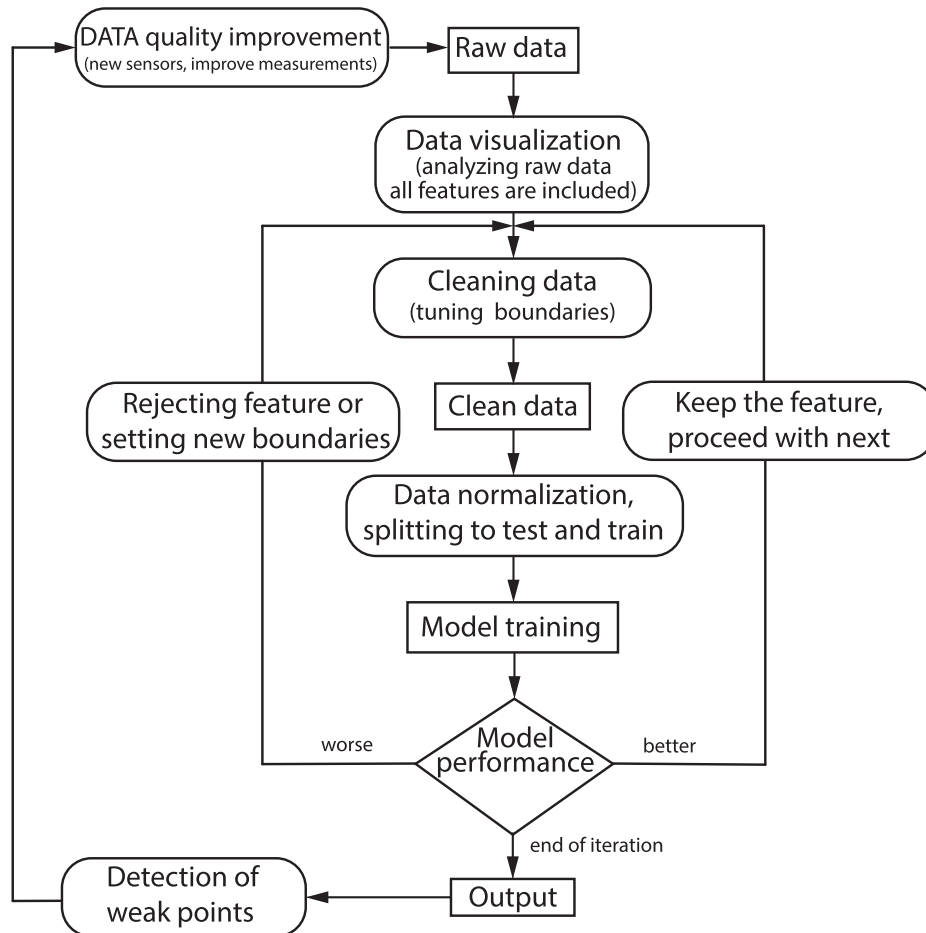


Fig. 2. Data-centric approach algorithm for improving data quality and model performance.

2. Experimental

2.1. Electric arc furnace process

This section will describe the melting process in an electric arc furnace, to present the diversity and complexity of all influencing parameters on energy consumption that generate nonlinearity. This detailed description will also connect research results with the design, capacity, and other characteristics of EAF, which will be discussed later, in Section 3.

2.1.1. Characteristics of electric arc furnace and process flow

An electric arc furnace has a nominal capacity of 60 t of liquid steel, with a transformer capacity of 60 MVA. The bath diameter (D) is 4100 mm, and its depth (H) is 920 mm. The EAF design ratio D/H is relatively low, compared to conventional design ratios: 4.8 – 6.2. This ratio is related to the heat of radiation and energy lost through cooling water, though encountering the reaction interface between slag and steel [31]. The electrodes are of an ultra-high power (UHP) type, controlled automatically by a separate control system. The electric arc's chaotic and dynamic nature is difficult to control as the arc length depends on nonlinear loads, which cause irregularities such as voltage dips, harmonic distortion, and flickering [32]. The typical distribution of energy consumption is shown in Fig. 1.

The total input energy consists of electrical and chemical energy, about half of which is used to melt waste material into steel. Dust and gas carry out about 25–30% of the input energy (Fig. 1). The gas in the furnace chamber contains mainly CO and CO₂, other components possibly being H₂, H₂O, N₂, CH₄, O₂. After leaving the furnace, CO is completely converted to CO₂, releasing a significant amount of heat. The released heat provides an additional energy source, which can be used to preheat the scrap, or improve the extraction of valuable metals from the dust, such as Zn and Pb [37–40]. Dust is collected in the dust filter system.

After refining and pouring liquid steel into the ladle, it is further processed in a ladle furnace (LF) to correct the chemical composition of liquid steel at the appropriate temperature. In the ladle furnace, liquid steel is treated to ensure the necessary level of alloying elements, to reduce the sulfur content, and to better control the finished product quality. The quality of liquid steel in the LF from the EAF is correlated with the previous treatment. Although indirect, the parameters in the LF provides information on the process in EAF.

2.1.2. Charging materials in the furnace

The loads fed into the furnace are steel scraps: a mix of different qualities, densities, and surface-to-volume ratios. The variety of bulk densities and compositions of waste leads to power quality problems and complex heat transfer through the materials, which affects the heat transfer to the steel and the overall energy consumption [32,34]. Alongside the energy consumption, the scrap composition and shape impacts the loss of steel and the amount of produced slag. The higher the content of the element that has to be removed, such as carbon, sulfur, phosphorus, silicon, the more prolonged the refining stage becomes, leading to longer operating times and higher energy consumption. Scrap charging in the basket follows a set process: scrap of a high density (such as castings) goes down, to enhance heat transfer from molten steel to scrap, while scrap of lower apparent density (such as shavings) is placed at the top of the basket, to facilitate melting with the electric arc via radiation. After each tapping, the molten steel – known as hot heel – is left, at typically 5–10% of capacity. The batch is then charged in the EAF without pre-heating. Although it has been proven that a significant amount of energy can be saved by preheating with waste heat, the

ecological benefits are doubtful due to the evaporation of dioxin or other organic pollutants [1,4,13,35]. Additional problems are presented by the complexity of the equipment and the surface oxidation of the charge. An innovative and more environmentally friendly approach uses phase change materials to recover waste heat [35]. After batching, a long arc is established to employ radiation heat. At this stage, refractories are protected with scrap; in later stages, a shorter electric arc is preferred to save refractory and cooling equipment, as well as saving heat consumption.

The loading basket's weights are 20–30 t, 15–25 t, and 10–15 t. The loading time is about four minutes, and melting for each basket takes about 25, 15, 10 min, respectively. The mean tap to tap time (TTT) is about 50 min. Typical operation, or the batch processing recipe, for one TTT involves separate stages: loading the scrap on the previous hot heel, melting, the second loading, melting, the third loading, melting, refining, adjusting the temperature with a reduction, and tapping. There are many possible ways to schedule the batch processing in EAF, and this directly impacts TTT and, consequently, energy consumption. As the number and mode of loading increases the waiting time, the TTT also increases, which causes the furnace to cool-down, and requires more intense energy for reheating. The literature states that this problem can be avoided by using path constraint modifications [33]. The typical downtimes, or delays, during the melting process are mainly caused when preparing the pouring hole and other related forms of maintenance, irregularities, or unusual furnace operations, as well as chemical and temperature analysis.

2.1.3. Additional process materials

Natural gas, oxygen, and carbon (coke) are chemical contributions to heat that influence the overall energy consumption (Fig. 1). Natural gas, as an additional heat source, is manually injected as needed, guided by experience. It is injected via four burners, each of which has one central oxygen injection, surrounded by holes for natural gas. Natural gas increases the energy intensity in EAF, avoids cold spots, and accelerates the scrap melting time [36]. Fine coke and lime are injected through one distinct injector. Portions of injected carbon enter the molten metal, while the rest goes to the slag, depending on the angle of injection and the particle size distribution. Coarse coke is added along with other materials in the basket, so it floats on the slag at a level depending on the type and quality of coke (or brown coal). Throughout the refining stage, oxygen is blown into a liquid bath; it reacts with carbon, sulfur, phosphorus, aluminum, silicon, manganese, and partly with iron, releasing energy from exothermic reactions, which raises the bath temperature. The amount of injected oxygen controls the chemical composition of produced liquid steel and affects the final liquid steel temperature. After the refining stage, aluminum is added to remove residual oxygen. This reaction is exothermic, but it contributes very little to the total energy, due to the small amounts involved.

Carbon, lime, and dolomitic lime are used to correct slag chemical composition and to make it foam. The slag foaming height is essentially related to the slag's basicity, viscosity, and surface tension. Slag in EAF has multiple contributions, such as: controlling oxidation and reducing the reactions of steel, carbon, and impurities; protecting refractories, cooling panels and equipment from aggressive heat radiation from the electric arc and chemical reactions; as well as enhancing power transfer from the electric arc to steel. All these requirements are met if the slag is foaming to a certain degree – the reaction between oxygen and carbon forms carbon-monoxide bubbles, which creates turbulence and leads to the foaming of the slag. Consequently, other than chemical energy contribution, intensive stirring by injected oxygen and carbon creates better heat transfer.

Furthermore, the amount of oxygen and strategies of carbon

injection influence iron oxides in the slag, resulting in the loss of metallic iron. The excess carbon in the slag reacts with the ferrous oxide, producing metallic iron, while extra oxygen causes oxidation and loss of metallic iron. Nitrogen is injected through the furnace's bottom aperture, to enhance mass transfer in the molten metal zone.

2.2. Data management

The dataflow management used in this research is shown in Fig. 2. Data quality is critical for improving the performance of machine learning models; a model based on invalid data will result in inaccuracies. The optimization process begins by understanding the process parameters, and identifying any anomalies or outliers in the raw data. Domain or expert knowledge with data visualization is essential for initially setting the boundaries of selected features. Fig. A.1 shows a correlation of some selected parameters after cleaning outliers. Plotting this data allows for quick detection of outliers (final boundary outliers are presented in Table A1). Before setting the boundaries, missing data (NaN) was replaced with zeros, meaning that the feature was not present in the melting operation. Zeros are only meaningful for features used periodically in the EAF process; for features such as 'Natural gas, Nm3', zeros mean that natural gas was not used. This is an error in the dataset, hence, those rows were dropped. Although these rejections have led to a lot of data being removed from the dataset, they are significant for improving the model's performance. After the initial set of constraints, the performance of the model is compared with that without any constraint: if the performance is higher, the iteration process continues. A new boundary is set, and the performance is compared to the previous one. This achieves the best results according to the metrics used. Features were selected as follows: first, all features were included, excluding 'Time, tap to tap' (which is redundant, as it is the sum of downtime and operating time features) and the main target. Next, the features were excluded one by one, by sorting through the importance of each method's feature permutation (Fig. 2). If the feature permutation was small for all methods, the feature was dropped. In the end, 24 features were selected for training, as they proved to be the most influential. In Fig. 2, the conceptual and further improvement of the data is presented by detecting any weak points in it, which is shown in the results section. The second stage of data improvement – improving data quality – refers to the installation of new sensors and improved measurements, which is beyond the scope of this study.

After cleaning and selecting the features, the data was normalized (using min-max normalization) and randomly divided into training and test sets, with the test set size at approximately 10%. In this way, the performance of the models was based on actual industrial samples, with a relatively sufficient number of samples.

Data was collected over the course of five years (from 2015 to 2020), with 14,836 meltings (or batches) and 42 features (process parameters). Data was collected in an Excel file (semi-automatic – human error here could be significant), each day in one sheet. Each sheet includes a section with comments on maintenance, irregularities, or unusual furnace operation, such as: changing an electrode, cleaning the furnace shell, problems with the carbon injection machine, cleaning the carbon injector, or pressing scrap after charging. Finally, after data cleaning, the training set had 10,521 entries, and the test set had 1169 entries.

To gain different perspectives on energy consumption, the machine learning models for targets were tested: 'Consumption EAF' (electricity consumption per tonne of produced steel, kWh/t), 'EAF operating time' (the time, in minutes, when electricity is on), and 'Yield' (percentage ratio between the mass of billets and the mass of the metallic charge).

2.3. Machine learning models

The most relevant regression techniques for machine learning analysis were used: Support vector machine, Random Forest, XGBoost, and Artificial Neural Network. The performance of each model prediction was evaluated using: coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE) metrics, maximum absolute error (MAX), and mean absolute percentage error (MAPE).

The transparency of quantitative relations between the process parameters and targets (i.e., explaining the target prediction by computing each feature's contribution to the prediction) is achieved using a partial dependence plot (PDP) and SHAP (SHapley Additive exPlanations) [41,42].

2.3.1. Support vector machine

The support vector machine (SVM) method is often used for regression tasks, as it is known for its excellent generalization capability, with high prediction accuracy [43]. Furthermore, in the literature [44], it is proved that the SVM with radial basis function kernel (RBF) is the universal approximator of arbitrary functions on a compact set, i.e., it can approximate any continuous function on a compact set with arbitrary accuracy.

In general, the RBF kernel is a reasonable first choice as, unlike a linear kernel, it can handle the possibility of a nonlinear relationship between target and attributes. On the other hand, the polynomial kernel has more hyperparameters than the RBF kernel, which therefore has fewer numerical difficulties.

There are two parameters for the SVM with RBF kernel: one defines the inverse of the radius of influence of samples, while the other behaves as a regularization parameter. As it is unknown beforehand which values are best for a given problem, a parameter search must be carried out. In this study, the best choice is a simple grid search since there are only two parameters, and the computational time required to find suitable parameters via grid search is not much more than that of more advanced methods.

The implementation of SVM from the scikit-learn library was used [45].

2.3.2. Random forest

Ensemble learning methods combine multiple machine learning algorithms and provide a prediction based on these. Specifically, the random forest (RF) is a regressor (or a classifier) that fits a number of decision trees on various sub-samples of the dataset, and uses averaging to improve the prediction and control over-fitting [28]. The decision trees are created by drawing a subset of training samples through replacement (a bagging approach). About two-thirds of the samples are used to train the trees, with the remaining one-third used in an internal cross-validation technique, to estimate how well the resulting RF model performs. Here, the algorithm creates trees with high variance and low bias by growing a forest up to a parameter number of trees. The RF predictor is formed by taking the average of all decision trees produced.

The RF involves several hyperparameters for controlling the forest structure (e.g., the number of trees), as well as its level of randomness and the maintenance of each decision tree's structure (e.g., the minimal size a node should have to be split). In most cases, RF works reasonably well with the hyperparameters' default values. However, tuning these hyperparameters can improve its performance [46]. A random search was used to tune the RF hyperparameters [47], because they are not equally important, and random searches over the same domain are able to find as good or even better models within a small fraction of the computation time than those obtained via grid search. The implementation of RF from the scikit-learn library was used [45].

2.3.3. XGBoost

Gradient boosting introduced by Friedman [48] has been particularly successful when applied to tree models. Boosting is one approach to building ensemble learning models. One variant of tree boosting, known for its exceptional predictive capacity, is Extreme Gradient Boosting (or XGBoost) [29].

The main idea of this algorithm is that it builds decision trees one by one, so that each subsequent model (tree) is trained using the residuals of the previous tree, i.e., the new tree corrects the errors made by the previously trained tree.

Like the Random Forest model, the XGBoost involves several hyperparameters for controlling XGBoost itself and controlling the decision trees' structure. Again, a random search was used to tune hyperparameters, as mentioned in the previous section.

The original implementation of the XGBoost library was used [29].

2.3.4. Artificial neural network

This paper uses a feedforward artificial neural network (ANN), more commonly called multilayer perceptron, which is a powerful learning method with widespread applications in many fields [48]. It has been proven [49] that multilayer feedforward networks are universal approximators, i.e., ANN can approximate any measurable function to any desired degree of accuracy.

Although a grid search or random search can be used to tune hyperparameters or ANN architecture, it would take an unacceptably long time. Therefore, the ANN hyperparameters were manually tuned.

ANN is implemented using the TensorFlow library [50]. It has four hidden layers, each consisting of 2048 nodes with a rectified linear unit (ReLU) activation. For the loss function, the mean absolute percentage error was chosen. The validation set is randomly selected inside the training set, so as not to compromise the test set.

3. Results and discussion

3.1. Setting different targets and model performance

The primary intention of setting different targets is to show how it impacts the results, and to view the outcome from the context of the processes' energy efficiency.

3.1.1. Electrical energy consumption

The electrical energy consumption ranges from 320 to 550 kWh/t, which almost follows a normal distribution with a mean value of 409.2 kWh/t and a standard deviation of approximately 27 kWh/t. Although RF and XGBoost are the most accurate models in the training set, according to the metrics followed, the model SVM-rbf is still the best in the test set (see Table 2). The test set is more relevant for interpreting the results or predicting the target value because the model in the training set may be overfitted. It can be surmised that there is a slight overfitting in RF and XGBoost.

Table 2

Comparison of training and test results with different machine learning methods for energy consumption as a target.

Model	Metrics				
	MAE	R ²	RMSE	MAX	MAPE
<i>Training set</i>					
SVM – rbf	6.17	0.904	8.37	61.0	0.0150
RF	2.51	0.984	3.39	33.5	0.0061
XGBoost	1.89	0.991	2.55	13.8	0.0046
ANN	5.73	0.921	7.60	56.3	0.0152
<i>Test set</i>					
SVM – rbf	6.37	0.901	8.36	48.2	0.0155
RF	6.83	0.888	8.94	39.3	0.0165
XGBoost	6.63	0.894	8.64	42.0	0.0161
ANN	6.62	0.897	8.60	46.4	0.0160

*Abbreviations: Support vector machines regression (SVM), Random forest regressor (RF), XG-Boost regression (XGB), Artificial neural networks (ANN).

Fig. 3 shows how the models' predicted values follow actual values in the testing set. It is evident that predicting the models' values follows actual value lines. The correlation of all predicted and actual values for the ANN testing set shows a relatively narrow error interval (Fig. 4).

3.1.2. Operating time

The operating time ranges from 28 to 45 min, almost following a normal distribution with a mean of 37.6 min and a deviation of approximately 2.24 min. According to the results shown in Table 3, all models decreased metrics compared with electricity consumption as a target. The SVM-rbf method had the worst, while ANN the best performance, indicating that this method is more flexible than others. Again, overfitting is evident in RF and XGBoost.

3.1.3. The yield of metallic steel

The performance of all model metrics decreases for 'Yield' as a target (see Table 4). This trend's primary cause was that the data quality was interrupted by the amount of molten steel left for the next batch (hot heel). The quantity of hot heels varied from 5 to 10% of the furnace capacity. In practice, the operator controls the amount of hot heel, so it is hard to maintain the same amount. The quantity of liquid steel left in the furnace affects the quality of yield data. Moreover, the quantity of heat that passes to the charged scrap disturbs the data quality of total electricity consumption per tonne of steel production. For these reasons, the yield for some batches surpasses 100%. After cutting this outlier data, the dataset was reduced to 11,405 data (Fig. 5).

3.2. Feature importance metrics

3.2.1. Electricity consumption analysis

Because the target depends on many inter-related parameters, PDP and SHAP were used to visualize and find quantitative relations between process parameters and targets. These quantitative relations have a wide area of confidence, though trends provide valuable information for each feature. Target dependence from two-component (2D plots) shows further transparency, so it may be beneficial to develop further dimension analysis methods.

SHAP analysis shows which features have the most influential impact on electricity consumption, as depicted in Fig. 6. The quantity of produced steel is included in the equation of electricity production; thus, it is evident that this feature is at the top of the impact list. Natural gas and oxygen contribute to chemical energy, which places them high, as the most influential parameters. It should be noted that the amount of oxygen is not divided into oxygen spent on natural gas, and oxygen used in the refining stage, so its impact is twofold, as will be discussed later. Operating time and quantity of charged material (steel fracture) directly increases electricity consumption. Coke, FeMn, and FeSiMn contribute energy through exothermic reactions. A waste type called Pas/MSM has the most significant impact on electricity consumption, compared to other scrap material types.

Trendlines for features influencing electrical energy consumption are presented as the middle line with dots in the diagrams in Fig. 7. The deviations from the trendlines are quite broad, though they inform how each feature impacts the target. The feature impact is more independent of others if the lines around the trendline are denser. On the other hand, for a greater spread in deviation lines, energy consumption depends on the combined effect of several features. The results in Fig. 9 show a relative change in electricity consumption in kWh/t. Selected features have a value of 0 for minimum and 1 for maximum, the boundaries of which are presented in Table A1.

The quantity of produced steel is included in the electrical energy consumption (electricity consumption per tonne of billets), so a linear dependence with a narrow deviation area is expected. Considering the boundaries for billets (Table A1), about 6.4 kWh/t of electrical energy will be lost per tonne of produced steel, which deviates from the furnace's maximum capacity (Fig. 7). Consumption increases linearly from

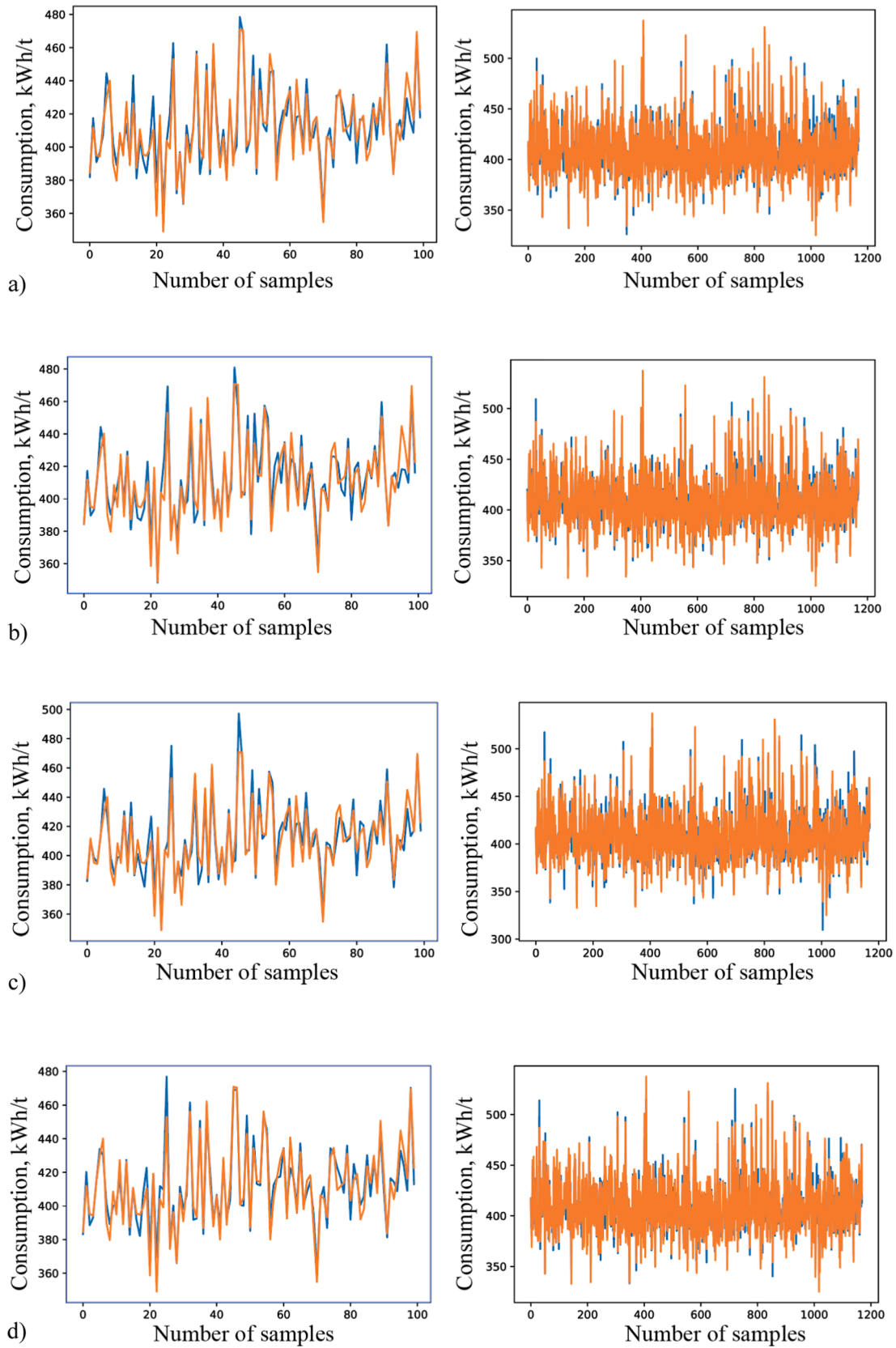


Fig. 3. Comparison of predicted (orange line) and actual values (blue line) for the first 100 and entire test set, using different machine learning methods: a) SVM-rbf b) RF, c) XGBoost, d) ANN. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

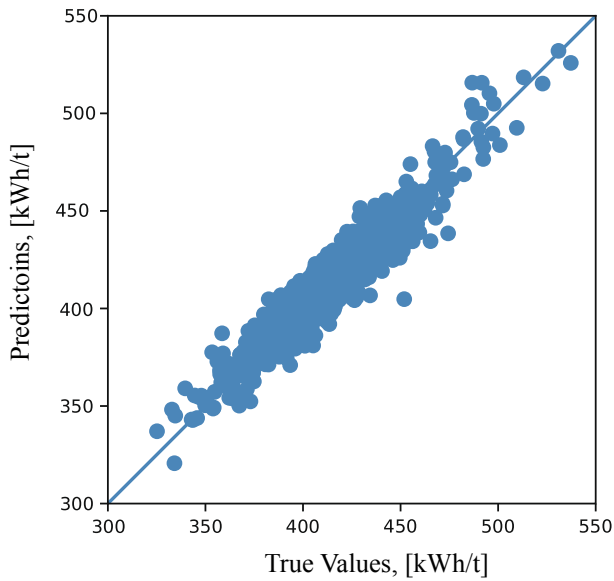


Fig. 4. Correlation between predicted and actual values, with ANN method.

Table 3
Performance metrics for different models and operating time as a target.

Model	Metrics				
	MAE	R ²	RMSE	MAX	MAPE
<i>Training set</i>					
SVM – rbf	0.663	0.816	0.961	10.0	0.0175
RF	0.254	0.974	0.358	3.43	0.0088
XGBoost	0.226	0.982	0.298	2.01	0.0059
ANN	0.600	0.844	0.880	10.0	0.0164
<i>Test set</i>					
SVM – rbf	0.739	0.795	1.033	7.51	0.0199
RF	0.711	0.832	0.966	5.42	0.0186
XGBoost	0.707	0.830	0.972	5.69	0.0189
ANN	0.672	0.847	0.922	6.14	0.0183

Table 4
Performance metrics for different models and yield of metallic steel as a target.

Model	Metrics				
	MAE	R ²	RMSE	MAX	MAPE
<i>Training set</i>					
SVM – rbf	1.03	0.852	1.45	10.5	0.0112
RF	0.632	0.943	0.902	8.12	0.0069
XGBoost	0.368	0.982	0.498	3.09	0.0041
ANN	1.07	0.855	1.43	15.0	0.0125
<i>Test set</i>					
SVM – rbf	1.14	0.819	1.57	12.0	0.0133
RF	1.12	0.833	1.51	12.6	0.0130
XGBoost	1.17	0.814	1.59	11.0	0.0141
ANN	1.15	0.839	1.52	7.50	0.0144

the amount of charged material (about 3.4 kWh/t per tonne of steel fractures), however, the deviation is obviously wider (Fig. 9). The linear increase of electricity consumption for charged material shows a very similar trend, with a linear dependence of electricity consumption on operating time (about 1.5 kWh/t per operating minute) (Fig. 9). In addition to the fact that more material requires more electricity in order to be melted, the quality and bulk density of scrap affects the heat transfer from radiation, and the conduction of heat from hot heel to charged material, and thus affects the time required for melting. Downtime trends close to zero because the two opposing effects are

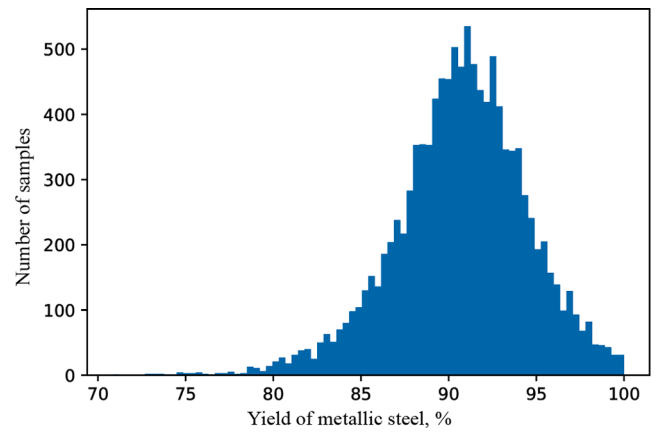


Fig. 5. Histogram of the yield of metallic steel; data showing yields greater than 100% were discarded.

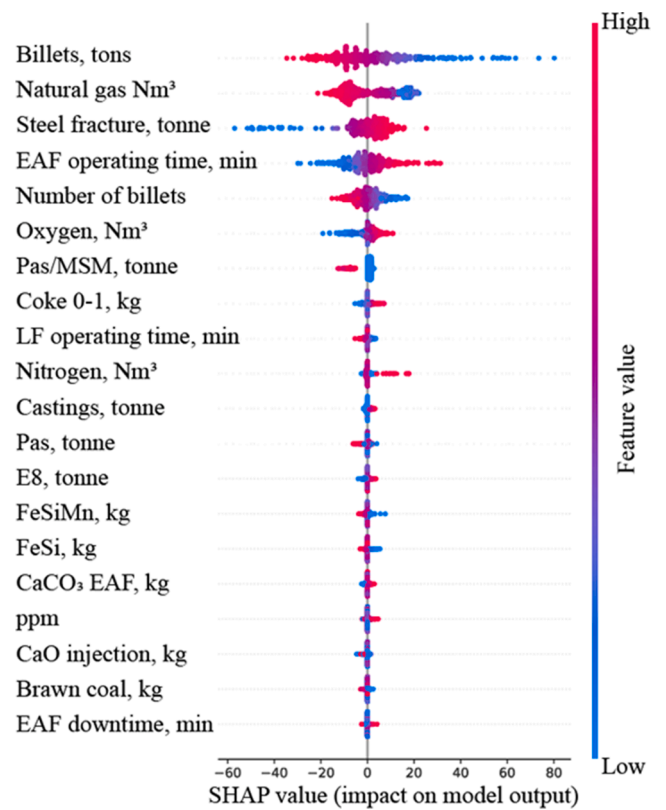


Fig. 6. SHAP values for electricity consumption target with ANN method, shows the most influential parameters.

conflicting: heat yield from chemical energy and heat loss from cooling water, off-gas and dust (Fig. 7). Separating the refining time from the downtime would give better insight into each feature's impact.

All the consumed oxygen is used for burning natural gas and reacting with carbon and other elements at the refining stage. Consequently, the amount of injected oxygen will affect its consumption by different mechanisms: it influences natural gas consumption, the exothermic reaction (mainly with carbon and alloying elements), and the oxidation of liquid steel. Only the latter increases electrical energy consumption because the steel oxidizes and turns into slag, leading to lower yield. As

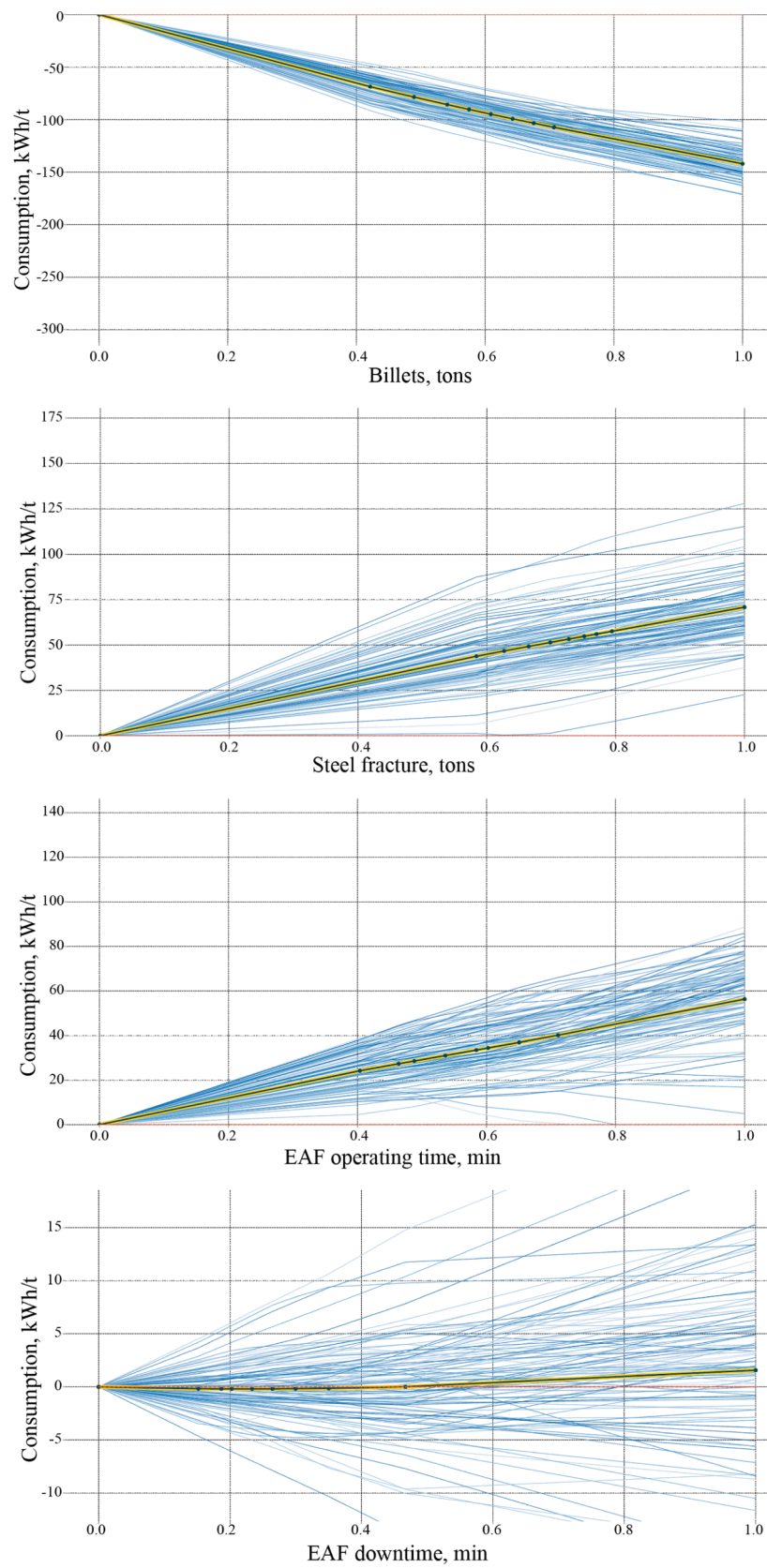


Fig. 7. ANN partial dependence plots for electricity consumption with produced steel, charged material, operating time, and downtime.

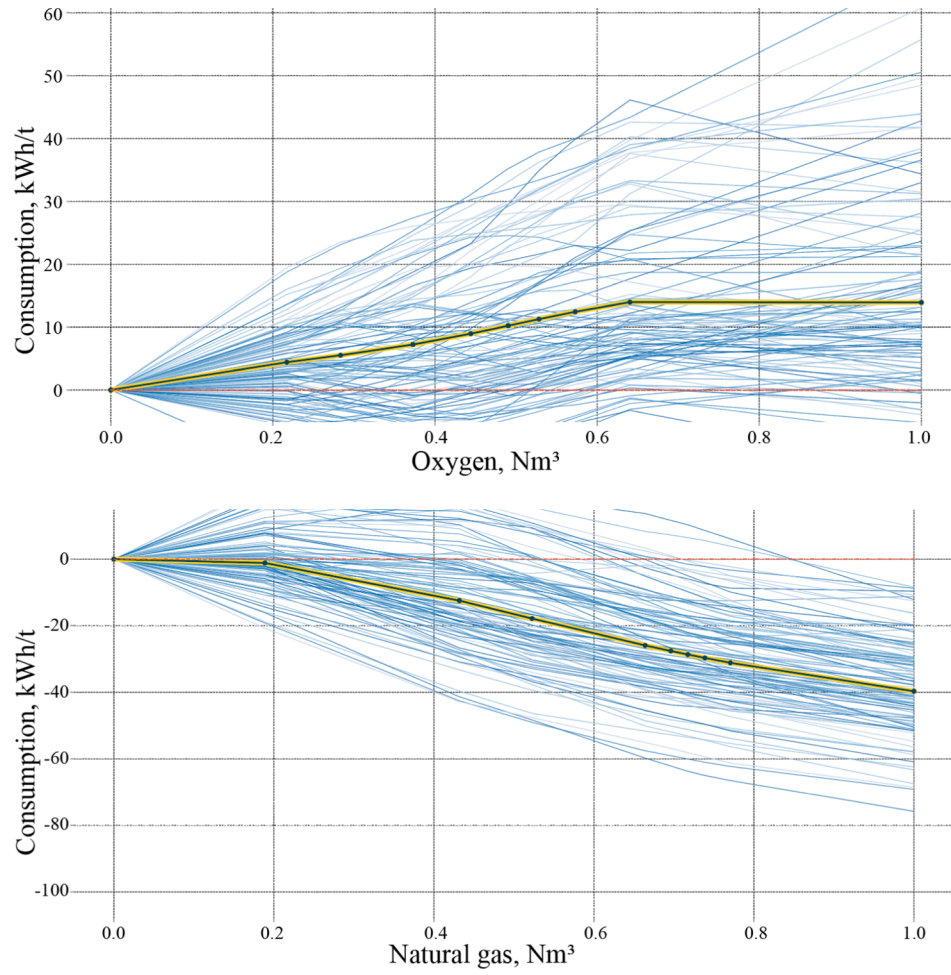


Fig. 8. ANN partial dependence plots for energy consumption with overall oxygen and natural gas consumption.

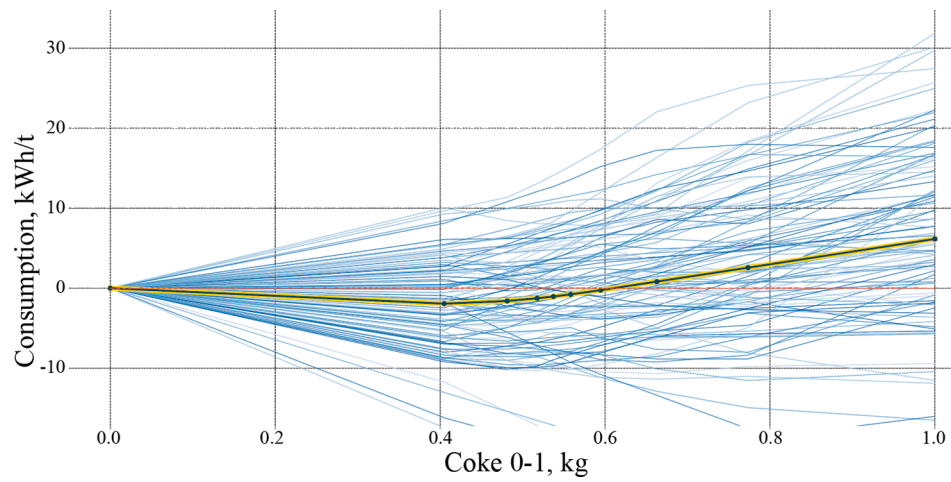


Fig. 9. ANN partial dependence plots for energy consumption with injected coke.

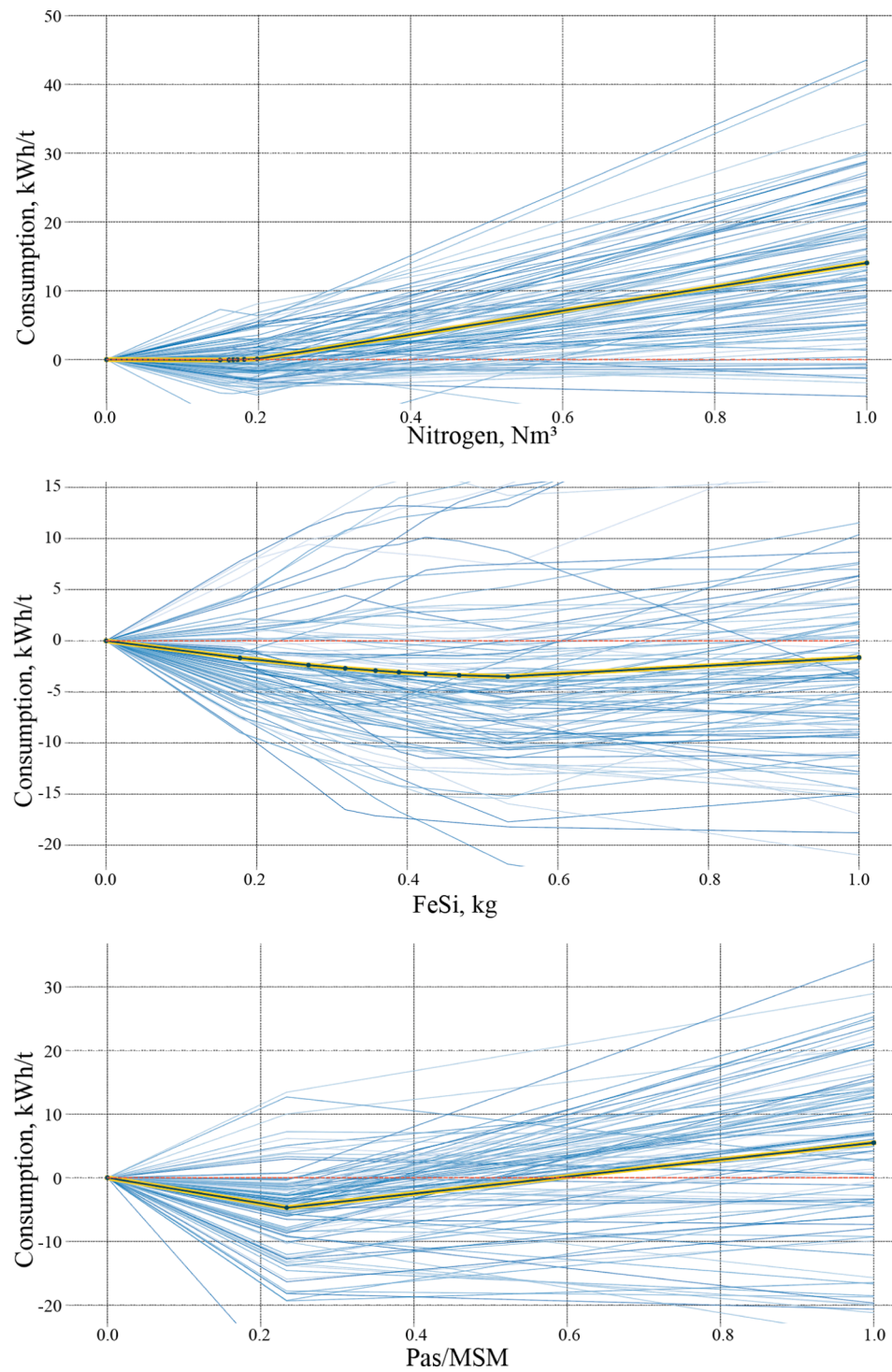


Fig. 10. ANN partial dependence plots for electricity consumption with nitrogen, FeSi, and Pas/MSM.

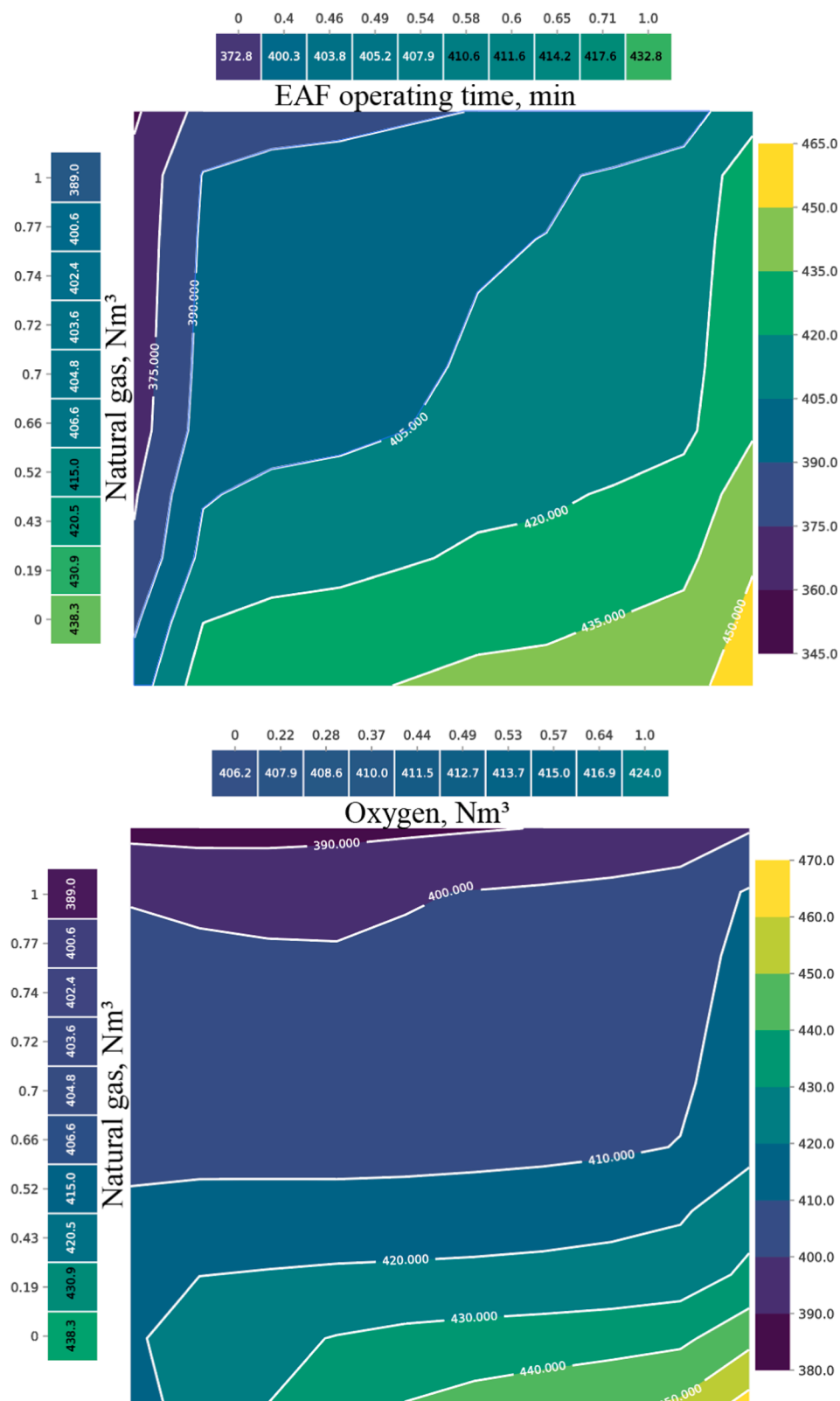


Fig. 11. ANN partial dependence of two features on electricity consumption, kWh/t.

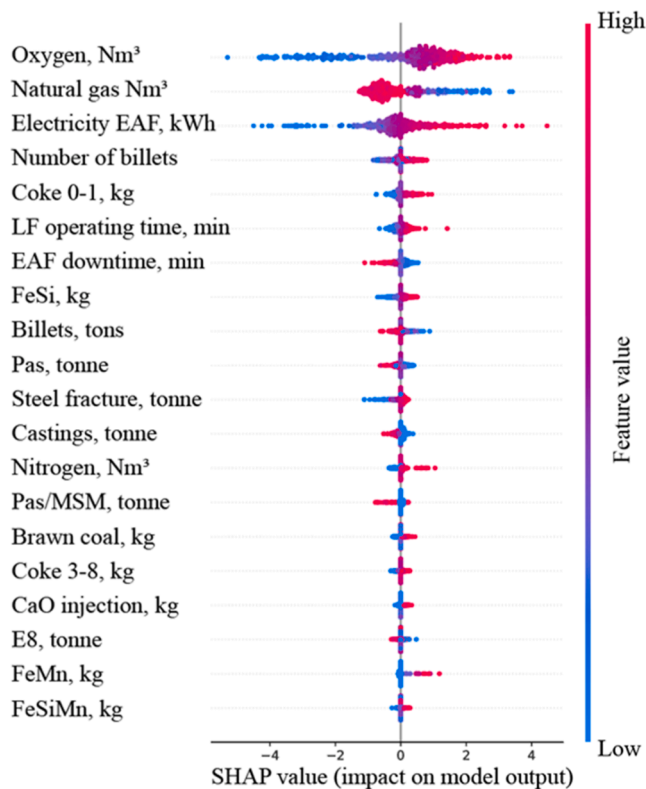


Fig. 12. SHAP values for operating time target with ANN method, shows the most influential parameters.

shown in Fig. 8, the steel loss prevails over the oxygen's chemical energy contribution. Separating the data on total oxygen consumption used for natural gas from the oxygen used for the refining stage would provide more information of the impact of oxygen on the overall process. It could also improve the performance of the model metrics. An increase in the quantity of natural gas leads to a decrease in electricity consumption after about 95 Nm³, and reaches a maximum of about 40 kWh/t at 550 Nm³ of natural gas consumption (Fig. 8 and Table A1). From these observations, it is evident that using a higher quantity of natural gas benefits electrical energy consumption. However, the heat loss caused by off-gas and the environmental impact of CO₂ emission are crucial variables in the overall equation of the optimum quantity of natural gas [51].

A portion of the fine injected coke dissolves in metal and, together with carbon from charged materials, contributes to chemical energy. Another portion of coke enters the slag and reacts with oxygen, causing the slag to foam and leading to high energy conversion efficiency, as explained in Section 2.1.3. The optimal amount of coke is between 300 and 470 kg (for Coke 0–1, Table A1 and Fig. 9); electrical consumption has a negative trend in this range. Nevertheless, higher coke rates negatively affect electrical consumption. This is perhaps counterintuitive as carbon should reduce ferrous oxides in the slag and increase steel yield, resulting in reduced electricity consumption. However, no information about the carbon content in steel scrap was available, so the ratio may be unsuitable, in turn leading to an unsuitable level of foaming, and increasing electricity consumption.

Nitrogen has a similar trendline to natural gas, yet in the opposite direction: over 20% of the maximum amount increases electrical

consumption (Fig. 10). This trend may be connected to the operating time: more nitrogen is consumed for longer operating times.

According to the SHAP analysis, the components that have the most impact on electricity consumption are Pas, Pas/MSM, and FeSi (Figs. 6, 12, and 14). Steel scrap categorized as Pas or Pas/MSM has a high carbon and manganese content; together with silicon from FeSi, these elements produce heat from exothermic reactions. The optimal amount of Pas/MSM is around 20% of the maximum (Table A.1 and Fig. 12). If the maximum amount rises to above 60%, Pas/MSM negatively influences electricity consumption, which could be attributed to bulk density and heat transfer from the electric arc to scrap.

Studying the impact of two components on the target provides more transparent information on the electricity consumption. According to Fig. 11, higher amounts of natural gas open the space for a longer operating time, maintaining the same electricity consumption. Less natural gas can be used with an average oxygen consumption (from 30 to 50% of the maximum amount). Still, a higher oxygen quantity leads to steel oxidation, which in turn leads to lower yield and electricity consumption, as shown in Figs. 7, 11, and 15.

3.2.2. Operating time analysis

Fig. 12 shows the SHAP analysis, which suggests that oxygen consumption is the most influential feature for operating time, even more so than electricity. This trend is counterintuitive, though it will be shown later that combining oxygen consumption with other features puts it in this first position. In this analysis, 'Electricity EAF' refers to electricity consumption per heat in kWh, a less relative parameter than electricity consumption per tonne of produced steel. Natural gas and coke contribute to chemical energy in order to melt the steel scrap, leading to a loss of material, thereby influencing operating time.

Reducing the operating time benefits electricity consumption, as shown in Fig. 7. It was found that the amount of oxygen has the greatest influence on operating time, as shown in Fig. 13. As discussed in Section 3.2.1, only the total amount of oxygen was tracked, so this study will comment on the combined impact of oxygen that burns natural gas, as well as oxygen used for refining. According to Fig. 7, after 60% of the maximal oxygen boundary has been reached, the trendline plateaus. Similarly, after about 50–60% of maximal oxygen is reached, the value increases the operating time, as shown in Fig. 13. Furthermore, low amounts of oxygen prefer areas of short operating times for natural gas, coke, Pas, and electricity per heat consumption.

3.2.3. The yield of metallic steel analysis

Fig. 14 shows that the quantity of produced steel is in the prime position because it is included in the yield equation. Electricity consumption is the next most influential parameter, as it is directly related to processed material. Natural gas and coke consumption influences yield, along with chemical energy, preheating, and thin scrap burning; this also puts scrap type Pas/MSM, and Pas high on the list.

A ladle furnace is used for correcting chemical composition: for high content of an alloying element or carbon, the LF operating time will be longer. Interestingly, the ladle furnace operating time is relatively high for all targets. Consequently, those indirectly related features should be highlighted when developing statistical models.

To gain better insight into the yield, the impact of other parameters was examined. Natural gas with electricity consumption expands the yield area at low and high quantities, as shown in Fig. 15. However, at low natural gas consumption, the yield ranges from 92.5 to 97.5%, and for high consumption it ranges from 87.5 to 95.5% (Fig. 15). Low amounts of natural gas involve less combustion or oxidation of thin

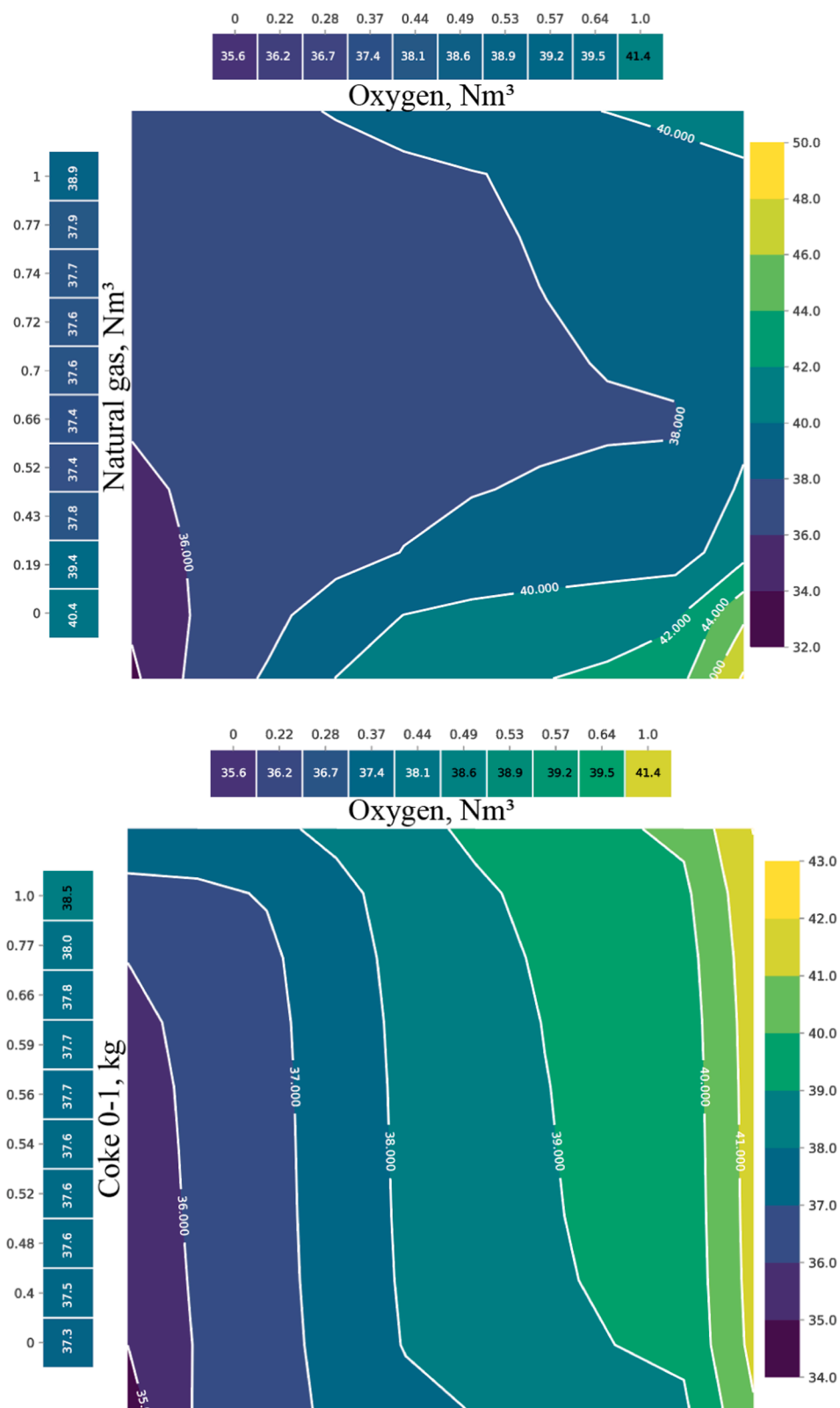


Fig. 13. ANN 2D partial dependence of impactful features with operating time as the target.

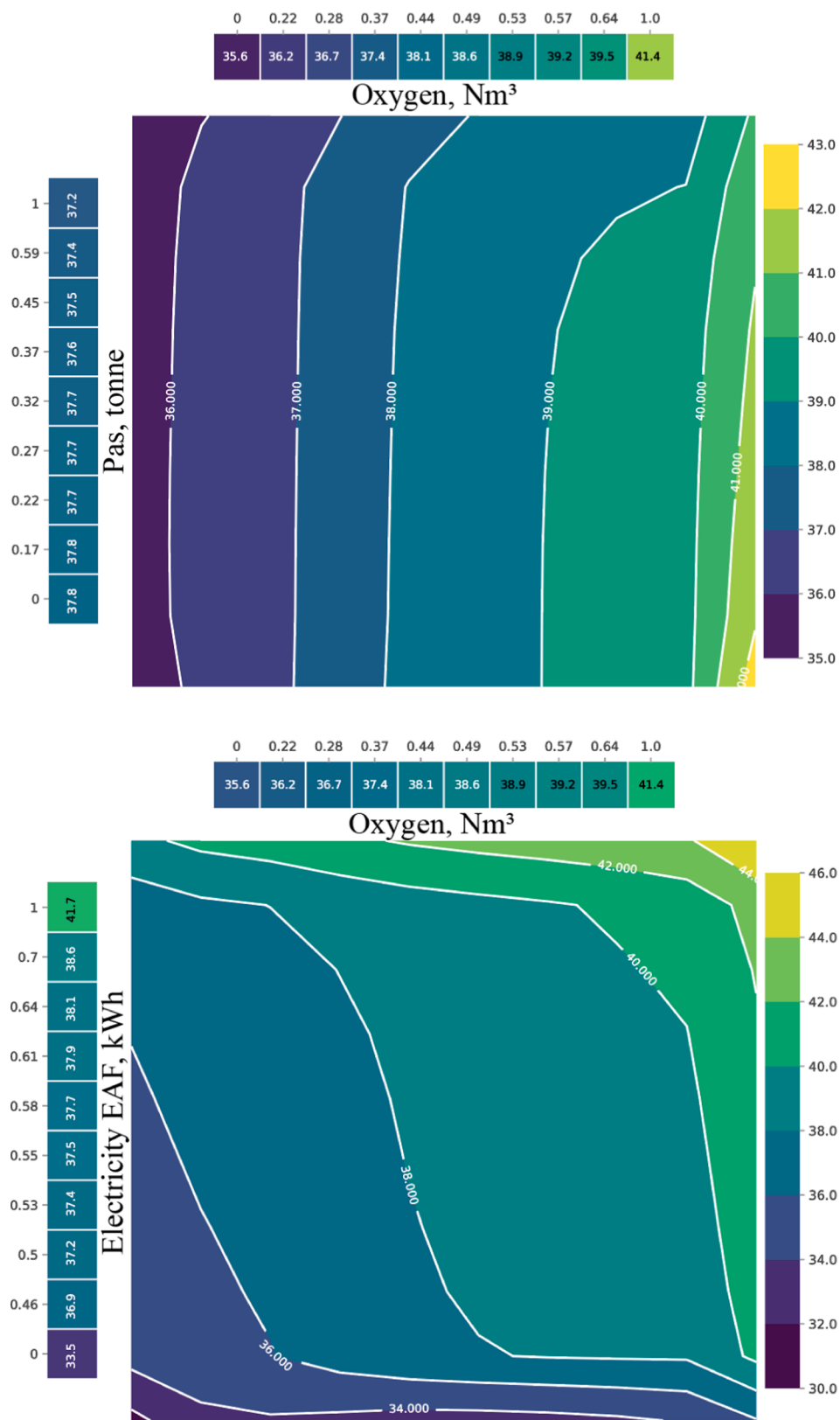


Fig. 13. (continued).

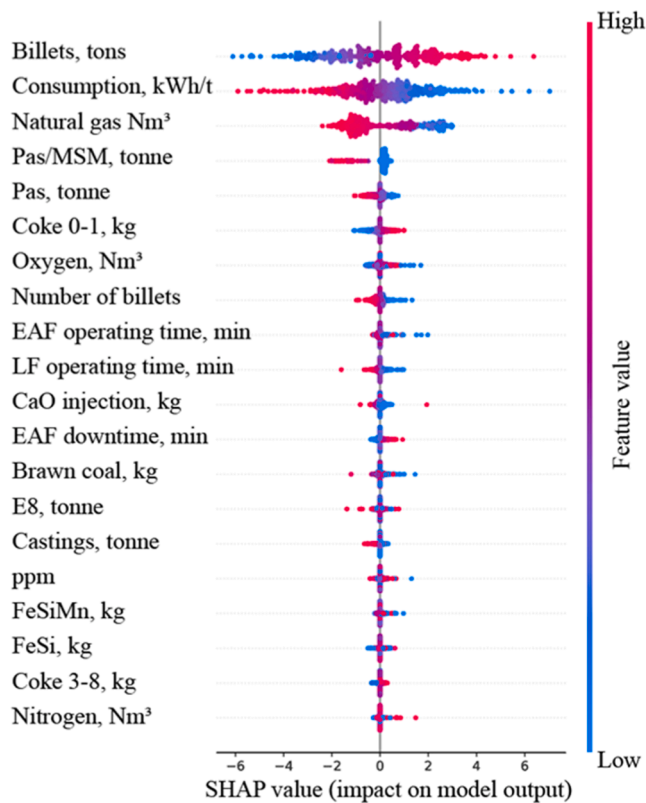


Fig. 14. SHAP values for Yield target with ANN method, shows the most influential parameters.

scrap sheets, while high quantities preserve electricity consumption. Additionally, at high and low operating times, the yield of steel areas expands. More efficient melting leads to higher yield over a shorter operating time. Conversely, for higher operating times, the impact of natural gas, together with electricity consumption, leads to yield of a wider range, as shown in Fig. 15.

3.3. Application of the presented approach

Complex industrial processes often rely on operator experience or engineering, which cannot grasp the impact of all parameters, causing significant energy losses. The development of complex physico-chemical software requires time and money, yet it fails to train the influence of all parameters without certain assumptions. The introduction of new parameters, or technology, into the process requires significant modification of the software. The main advantage of statistical models is being able to avoid the simplifications and assumptions which are present in real working conditions. The wider community strongly supports the machine learning methods presented in this research. This study has shown that there is a slight difference in the metric performance for different model strategies. Applying machine learning methods can simplify engineering in large systems by following the proposed approach. Building these kinds of online prediction software or systems in complex industrial processes is simple, flexible, and makes it easy to change and experiment with new features.

Data quality is at the core of the machine learning approach, so it is essential to find methods to improve it. With the proposed data-centric approach for data management through training and testing of the new

machine learning model, this study has shown that the data quality has significantly improved. Furthermore, boundaries can be set for each feature, and feature selection can be made using initial engineering knowledge, visualization tools, and model iteration. Cleaning data involves removing bad data and understanding where errors occur and how they impact the model quality.

Employing a machine learning model to control and optimize main parameters, such as oxygen or natural gas consumption, leads to less energy consumption and CO₂ emission of EAF, or similar processes [52]. Analysis of parameters through different targets can reveal hidden information about process parameters and energy consumption. A connection between some of these indirect parameters was found, as demonstrated via the ladle furnace operating time and selected targets. This influence of indirect parameters shows how the proposed method can be applied to complex systems, in which direct measurements are difficult to perform. Also, parameters that could more precisely be measured were identified, such as hot heel and oxygen consumption. The amount of hot heel (melted steel left in the furnace for the next batch) was not measured. It was found that the amount of hot heels was the main factor disturbing the quality of data and influencing the model performance. Therefore, it is essential to get high precision measurements to implement statistical models in real-time conditions and to track more parameters.

Furthermore, the total oxygen consumption should be divided into oxygen used for refining, and that used for natural gas. Equally, downtime should be divided into refining time, and actual downtime. Only small investments in more sensors or logistics for collecting data would improve reliability in statistical methods. Future research could be done on integrating these models into existing or new control system hardware. Experimenting in industrial conditions with complex systems becomes more agile when using cheap sensors, and the expertise of engineers could be more process-related.

One application problem is the relatively small amount of data that was collected over an extended period of time. Nevertheless, the data set showed good reliability of about 11,000 entries for SVM-rbf and ANN models, and some overfitting for RF and XGboost. This problem of overfitting can easily be overcome if there was more data available. Furthermore, various furnace designs and conditions would give different correlations between the same features, which would limit the use of the same software to similar processes. These problems can be solved by gathering valid data from reputable communities or organizations (in this case, the World Steel Association, for example), guaranteeing consistent and responsible (e.g., encrypted to protect sensitive company data) data collection. Moreover, such an approach would provide insight into the process' design efficiency and its impact on energy consumption.

4. Conclusion

Using machine learning analysis, it was shown that it is possible to find a correlation between many processing parameters within nonlinear and complex systems, such as the process of melting steel waste in an electric arc furnace. It was found that support vector machines and artificial neural networks are the most accurate, according to metrics. However, an artificial neural network shows greater flexibility when targets are changed. The coefficient of determination (R^2) was about 0.9, and a comparison of predicted and actual value lines show an excellent fit in the test set for all regression analysis models. Data quality is critical for achieving good machine learning model performance. About 21% of input data was dropped due to missing or irrational data

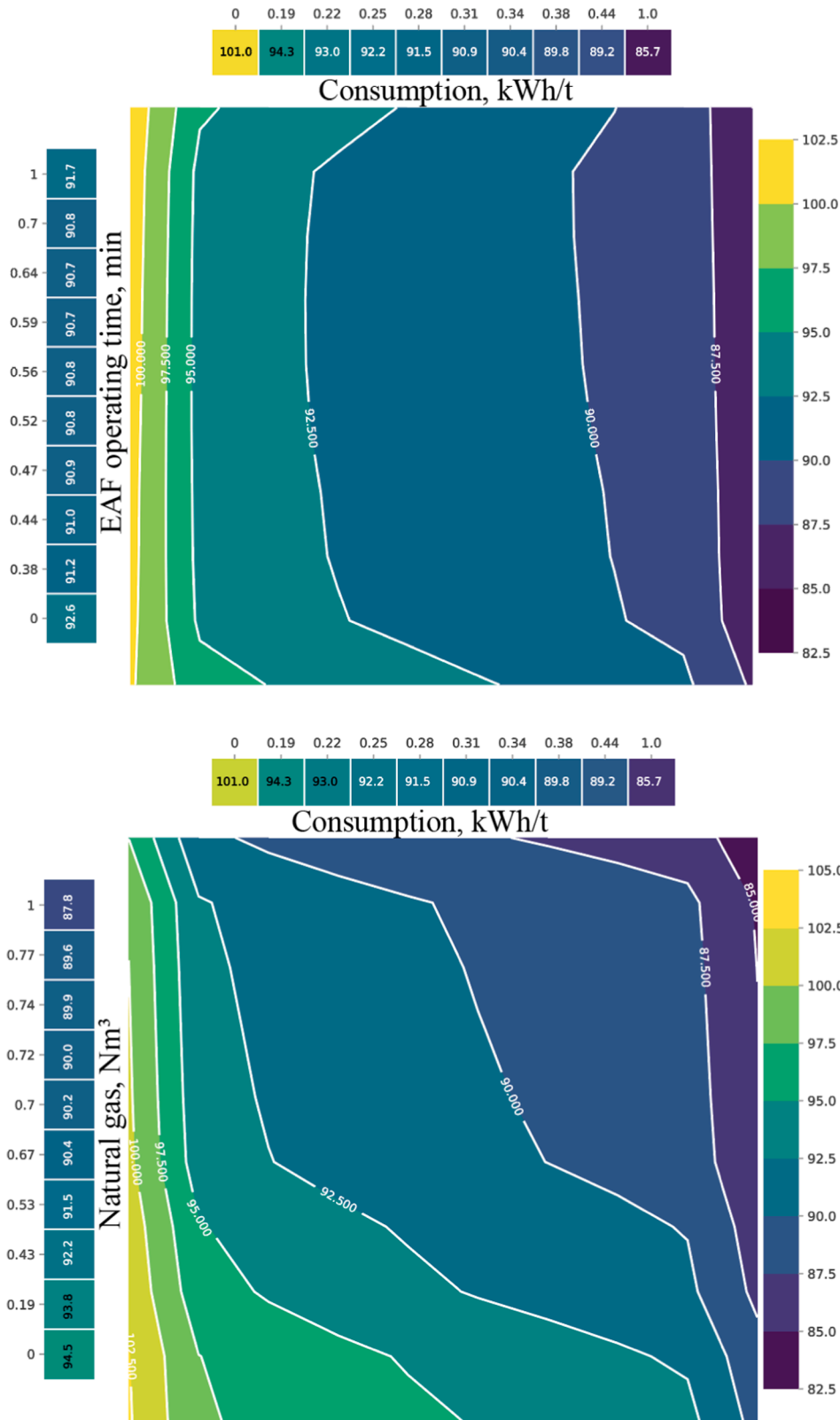


Fig. 15. ANN 2D partial dependence of impactful features for the yield of metallic steel, in %.

Table A1

Features labels, boundaries after cleaning, total number, and the number of dropped data.

Feature	Lower limit	Higher limit	Total number (without zeros)	Number of dropped data
Time, tap to tap, min (sum of operating and downtime)	40.0	66.0	14,836	547
EAF operating time, min (time with electricity on)	0.0	45.0	14,836	230
EAF downtime, min (time with electricity off)	0.0	30.0	14,836	7
LF operating time, min (refers to ladle furnace)	6.0	45.0	14,835	130
LF time, min (refers to ladle furnace)	3.0	45.0	14,835	306
Oxygen, Nm ³ (the amount of oxygen that is blown, expressed in normal m ³)	1050.0	2400.0	14,836	91
Natural gas, Nm ³ (the quantity of natural gas injected)	80.0	550.0	14,030	693
Nitrogen, Nm ³ (the amount of nitrogen gas blown)	0.0	350.	10,862	84
Electricity EAF kWh (total electricity consumption in EAF per one heat, kWh)	19000.0	29000.0	14,836	27
Electricity LF kWh (total electricity consumption in LF)	800.0	3800.0	14,835	164
Coke 0–1, kg (quantity of injected coke, with granulometric composition below 1 mm)	20.0	800.0	14,826	140
Coke 3–8, kg (quantity of charged coke, with granulometric composition between 3 and 8 mm)	0.0	200.0	14,673	5
Brown coal, kg (quantity of brown coal)	0.0	900.0	9848	35
CaCO ₃ LF kg (quantity of charged lime in ladle furnace)	100.0	1100.0	14,822	32
FeSiMn, kg (quantity of charged ferro silicon manganese alloy in EAF)	0.0	1200.0	14,815	12
FeSi, kg (quantity of charged ferro silicon alloy in EAF)	0.0	450.0	14,741	11
Al, kg (quantity of charged aluminum in EAF)	0.0	40.0	7786	65
CaCO ₃ EAF, kg (quantity of charged lime in EAF)	150.0	2000.0	14,813	23
CaO injection, kg (quantity of injected fine CaO)	0.0	750.0	8740	45
ppm (quantity of oxygen in liquid steel, expressed in ppm, measured with a probe)	110.0	1500.0	14,820	24
Number of baskets (number baskets used for charging the scrap)	2.0	4	14,834	1
Billets, tonne (the quantity casted billets, approximately same as produced steel)	47.0	70.0	14,833	25
Steel fracture, tonne (total quantity of charged scrap)	52.0	74.0	14,836	12
Shavings, tonne	1.0	14.0	14,530	212
Mix, tonne	0.0	18.0	14,435	88

Table A1 (continued)

Feature	Lower limit	Higher limit	Total number (without zeros)	Number of dropped data
E1, tonne (old thick steel scrap)	2.0	35.0	14,653	59
E8, tonne (thin new production steel scrap)	0.0	28.0	4984	91
E3, tonne (thick new production steel scrap)	0.0	25.0	14,342	4
Cast iron, tonne	0.0	8.0	10,043	21
Coke 10–50, kg (quantity of charged coke, with granulometric composition between 10 and 50 mm)	0.0	850.0	1187	1
FeMn, kg (quantity of charged ferro manganese alloy in EAF)	0.0	600.0	819	1
FeSi briquet, kg (quantity of charged ferro silicon alloy in form of briquet)	0.0	120.0	92	2
MSM briquet, kg (quantity of charged ore briquettes in EAF)	0.0	2000.0	106	1
Castings, tonne	0.0	6.0	4984	10
Schroeders, tonne	0.0	13.0	6931	7
Briquet Shavings, tonne	0.0	6.0	169	1
FeSiC, kg (quantity of charged ferro silicon carbon alloy in EAF)	0.0	212.0	725	0
Pas, tonne (old thick steel scrap, above 6 mm)	0.0	54.2	13,021	0
Pas/MSM, tonne	0.0	58.1	1624	0

caused by human error or some irregularities. A dataflow management algorithm was proposed, which includes domain knowledge and visualization for initial-boundary setting, as well as statistical analysis for further polishing. After cleaning the data, 24 features were selected as optimal for training, by iterating through the models and using a partial dependence plot and SHAP methods. By carrying out an in-depth analysis of each feature, the impact of each on electricity consumption was quantified. Weak points were also identified, such as a lack of precise measurements for oxygen consumption and hot heel. A small capital investment can eliminate these shortcomings and increase the machine learning models' performance, giving deeper insight into the process. Using the proposed approach, prediction software can be built to work in complex industrial systems, for flexible and straightforward experimentation. Such an approach would give experts in the field more freedom to use their knowledge, to optimize complex processes, and to reduce energy consumption.

CRediT authorship contribution statement

Vaso Manojlović: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft. **Željko Kamberović:** Resources, Supervision. **Marija Korać:** Project administration, Writing – review & editing. **Milan Dotlić:** Validation, Visualization, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

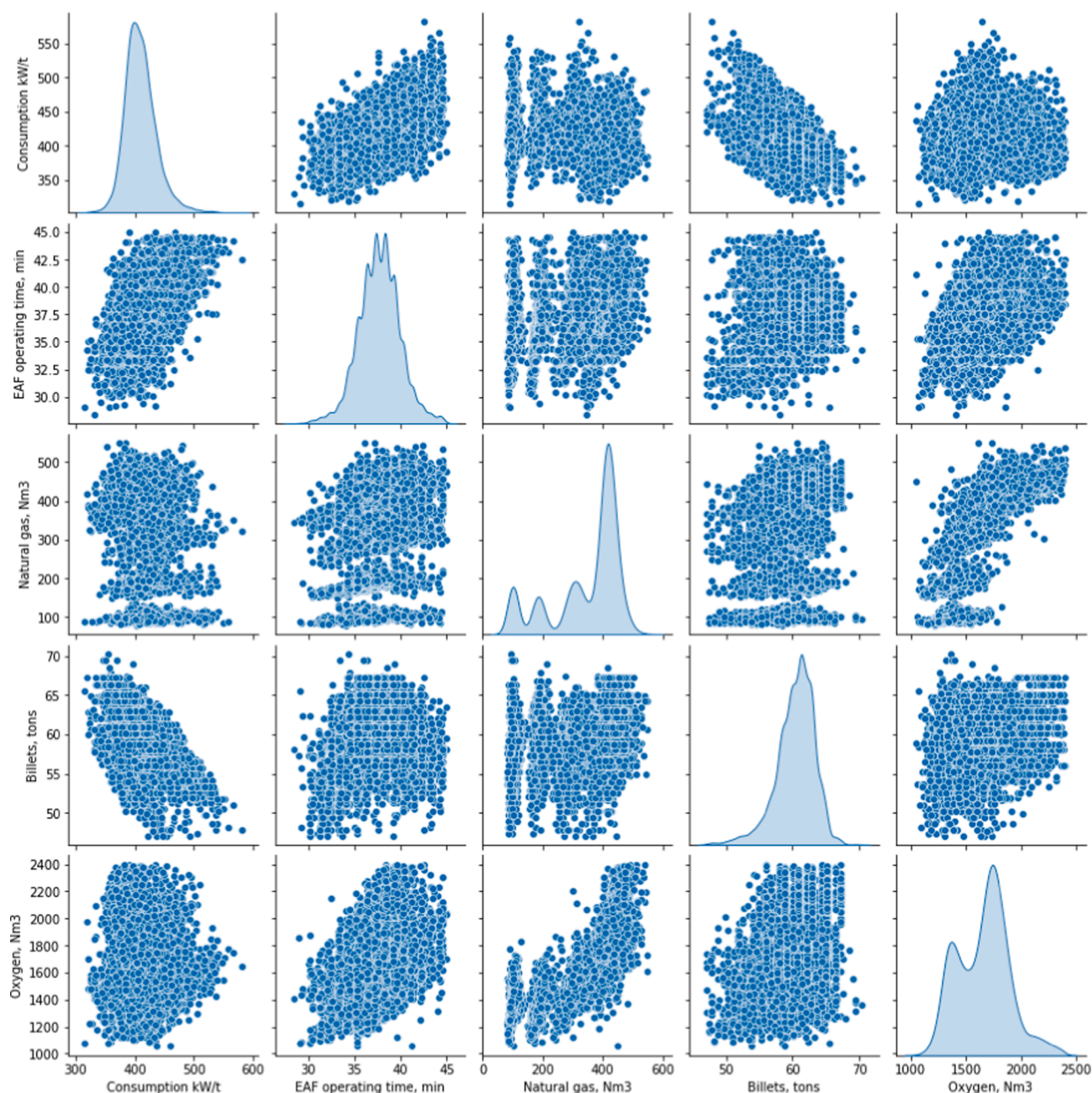


Fig. A1. Pairwise scatter plots show a correlation of some selected parameters after data cleaning.

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Appendix

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